

WEST TEXAS A&M UNIVERSITY
Department of Media Communication

Independent Study Proposal

AI-Powered Podcast Production Automation Pipeline

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Project Overview

This independent study proposes the design and development of an end-to-end AI-powered podcast production pipeline. The system will autonomously research a topic, generate a multi-host script, synthesize realistic voices, apply professional audio post-processing, and export a broadcast-ready episode — all with minimal human intervention. The project bridges mass communications theory with practical AI engineering, culminating in a working software system..

Learning Objectives

- Understand and apply audio production theory, including sound design, pacing, and narrative structure in a broadcast context.
- Design and implement a multi-agent AI pipeline using LLM orchestration frameworks (e.g., LangChain, CrewAI, or custom agentic loops).
- Integrate text-to-speech synthesis APIs (e.g., ElevenLabs, Resemble.ai) to produce distinct, expressive host voices.
- Apply gatekeeping and agenda-setting theory to the AI editorial decision-making process.

- Evaluate the ethical implications of AI-generated audio media, including authenticity, transparency, and audience trust.
- Produce a polished academic paper demonstrating mastery of both mass communications scholarship and technical implementation.

Technical Architecture

Stage 1 — Topic Research Agent

An agentic LLM (GPT-4 class or equivalent) uses web search tools to gather current information on a user-supplied topic. The agent synthesizes sources, identifies key angles, and outputs a structured research brief. Technologies: LangChain or CrewAI, web search API (Tavily or Brave), vector store for source deduplication.

Stage 2 — Script Generation

A second LLM call transforms the research brief into a two-host conversational podcast script with a defined opening, segment structure, transitions, and closing. Prompt engineering is used to enforce tone, word count targets, and journalistic balance. Output: structured JSON with speaker turns and metadata.

Stage 3 — Voice Synthesis & Audio Assembly

Each speaker turn is passed to a TTS API with a unique voice profile. Audio segments are exported and assembled using a Python audio library (pydub or ffmpeg). Background music, intro/outro stings, and compression/normalization are applied programmatically. Output: .mp3 episode file.

Stage 4 — Metadata & Distribution Package

The system generates episode metadata (title, description, tags, chapter markers) and packages the episode for RSS feed compatibility. Optional: automated upload to a hosting platform via API.

Workload Breakdown

Task	Phase	Deliverable
1.1	Literature Review	Annotated bibliography: AI in media, podcast production, TTS ethics
1.2	System Design	Architecture diagram, tech stack selection, API keys secured
1.3	Research Agent	Working topic ingestion and summarization module
1.4	Script Engine	Script generator with configurable hosts, tone, and length
1.5	Voice & Audio	TTS integration, audio assembly, and post-processing pipeline

1.6	End-to-End Testing	1 full pilot episode produced and evaluated
1.7	Project Summary Paper Draft	Technical Summary draft covering theory, design, and findings
1.8	Final Polish & Presentation	Polished system, final paper, and faculty demo

Workload Distribution

Component	Weight	Description
Working Pipeline (Demo)	60%	Fully functional end-to-end system producing a listenable episode
Technical Documentation	10%	Documentation for all technical aspects of the system
Pilot Episodes	25%	Complete episodes demonstrating system quality and variety
Final Presentation	5%	Live faculty demo with Q&A

Proposed Resources & Tools

- LLM API access: OpenAI GPT-4 or Anthropic Claude (student account or WTAMU research access)
- TTS API: ElevenLabs starter plan (~\$5/month) or Resemble.ai
- Python 3.11+ environment with pydub, ffmpeg, langchain, and related libraries
- Web search API: Tavily (free tier) or Brave Search API
- Podcast hosting: Buzzsprout or Anchor (free tier) for distribution packaging
- Version control: GitHub repository for all source code
- Audio editing review: Audacity (free) for quality spot-checks

Academic Grounding

The project draws on foundational mass communications theory throughout. Gatekeeping theory (Shoemaker & Vos, 2009) informs how the research agent filters and prioritizes information. Agenda-setting theory (McCombs & Shaw, 1972) is applied to evaluate how AI editorial choices shape topic framing. Production workflow theory from audio journalism (McLeish & Link, 2016) guides the pipeline's structure.

Approvals & Signatures

Student Signature

Faculty Supervisor Signature

Date

Date